DOORS Software Requirements Specification

Data Orchestration and On Demand Reporting System

GePNIC Data Governance and Secure Integration Platform

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1 Introduction

1.1 Purpose

This Software Requirements Specification defines the required behaviour, interfaces, security controls, data rules and operational qualities of the Data Orchestration and On Demand Reporting System. It provides the common baseline for product governance, software development, security review, database provisioning, deployment, testing and acceptance. Each normative requirement has a stable identifier for traceability.

1.2 Product Scope

DOORS is a governed data bridge for the Government e Procurement platform operated by NIC. It connects distributed GePNIC database and document nodes to authorised human users and machine clients without exposing raw database credentials, direct database ports or unrestricted SQL. The platform supports governed read queries, versioned JSON writes, reports, controlled document delivery, API subscriptions, audit investigation and AI assisted operational guidance.

1.3 In Scope

- Administrative, external-user and machine-API request planes with fail-closed route separation.
- Human identity, CAPTCHA, password, MFA, VPN, IP and role-based access controls.
- Governed data-request, read-query, report, write-contract and document-service lifecycles.
- Agent registration, mapping, health, sandbox testing and destination verification.
- API-client credentials, public keys, IP allowlists, activation, service permissions, SDKs, Swagger and OpenAPI.
- Encrypted and signed machine requests, encrypted responses, replay protection and traceability.
- Versioned inbound JSON storage using DBA-provisioned PostgreSQL functions and latest-record views.
- Audit logs, execution logs, usage measurement, subscriptions, schedules, notices and troubleshooting.
- AIra retrieval-grounded assistance using governed operational knowledge and live activity tools.

1.4 Out of Scope

- Replacement of GePNIC transactional applications or their primary business logic.
- Direct end-user access to GePNIC database credentials or unrestricted database consoles.
- Execution of unapproved DDL or DML by live API requests.
- Automatic modification of destination schemas by the live write API.
- Automatic financial settlement or invoice payment for API subscriptions.
- Storage of API-client private keys or wallet passwords by DOORS.

1.5 Definitions and Conventions

The words shall and must indicate mandatory requirements. Should indicates a recommended requirement. Priority is classified as Must, Should or Could. Verification methods are Inspection, Test, Demonstration or Analysis. API and database examples are illustrative unless explicitly identified as fixed contracts.

2 Overall Description

2.1 Product Perspective

DOORS uses a central Master API and web user interface with remotely deployed Agent APIs. The Master API governs identities, approvals, API clients, templates, contracts, orchestration, auditing and AIra. Each Agent

connects to one governed GePNIC database and, where configured, document storage. Reverse proxies expose separate Admin, External and API hostnames and forward only the routes assigned to each plane.

2.2 Product Objectives

Objective	Required outcome
Least privilege	Every read, write and document action is limited by independently governed identity, client, service, Agent and data scopes.
Separation of duties	Proposers cannot approve their own governed artefacts; operational and security administration remain separate.
No direct exposure	Consumers do not receive database credentials, direct database access or arbitrary SQL capability.
Verifiability	Every material action produces traceable status, audit evidence and a stable trace identifier.
Interoperability	Approved integrations are delivered through documented HTTPS APIs, OpenAPI contracts and generated SDKs.
Controlled evolution	Query definitions, JSON contracts and API policies evolve through explicit versioned governance.

2.3 Assumptions and Dependencies

- NIC network, VPN, DNS, PKI, reverse proxy and firewall services are available and managed outside DOORS.
- A GePNIC DBA executes approved provisioning or migration SQL in each destination database.
- Each Agent has trusted, least-privilege connectivity to its mapped database and access to required file paths.
- System clocks are synchronised sufficiently for TOTP, signed-request freshness and audit correlation.
- PostgreSQL, Java, the web runtime and configured AI/vector services meet the supported deployment baseline.
- Application owners securely retain private keys and protect wallet and truststore passwords.

3 System Context and Architecture

3.1 Three Plane Architecture

Plane	Primary users	Permitted capabilities	Required isolation
Admin Plane	Administrators, Data Managers, Security Administrators and Developers	Governance, registries, approvals, infrastructure, auditing and administration	Reject external and machine-only routes
External Plane	External users and Data Viewers	Request submission, status tracking, authorised reports and external self-service	Reject administrative and gateway orchestration routes
API Plane	API users and machine clients	Gateway execution, writes, document operations, SDK-guided API use and scoped	Reject unrelated administrative routes and

Plane	Primary users	Permitted capabilities	Required isolation
		subscriber portal	enforce machine admission controls

The host, route and authenticated role shall all agree with the selected plane. Requests that cannot be assigned safely shall fail closed and shall not be rerouted to a more permissive plane.

3.2 Logical Components

Component	Responsibility
DOORS UI	Role-aware web interface for governance, reporting, onboarding, testing and audit investigation.
Master API	Authentication, authorisation, orchestration, governance, cryptography, SDK generation, auditing, AIra and policy enforcement.
Agent API	Controlled database query execution, write-function invocation, schema verification, sandbox testing and document access at the GePNIC node.
Master database	Users, governance metadata, client policies, mappings, approvals, subscriptions, audits and vector knowledge index.
GePNIC database	Existing read schema plus DBA-owned doors_write schema, approved write functions, history tables and latest-record views.
Reverse proxy	TLS termination and allowlisted routing for the three planes and Agent endpoints.
Vector and model services	Retrieval of approved knowledge chunks and controlled answer generation for AIra.

4 User Classes and Responsibilities

User class	Primary responsibilities	Key restrictions
Administrator	Platform configuration and operational administration	No implicit bypass of governance, client or data scope
Security Administrator	User lifecycle, role, VPN entitlement, password and MFA recovery	Does not perform routine data-query approval
Data Manager	Approvals, Agents, API clients, policies, subscriptions and audit investigation	Cannot self-approve a governed proposal
Developer	Creates parameterised read templates and tests against authorised sandboxes	SELECT-only; cannot approve own SQL
Data Viewer	Runs approved interactive reports and permitted exports	No SQL visibility or governance access
External User	Submits justified data requests and tracks their status	No report access until authorised
API User	Manages assigned integrations and uses approved SDKs and APIs	No administrative access outside assigned clients
GePNIC DBA	Installs approved write schemas/functions and applies grants	Uses approved SQL; live API does not grant DDL

User class	Primary responsibilities	Key restrictions
Machine Client	Calls assigned APIs using its API key, private identity and approved source IP	Cannot exceed mapped service, query, Agent or data scope

5 Functional Requirements

5.1 Identity and Session Management

Requirement ID	Requirement	Priority	Verification
FR-AUTH-001	The system shall require a valid CAPTCHA before processing an interactive username and password authentication attempt.	Must	Test
FR-AUTH-002	The system shall enforce configurable password composition, length, reset and forced-change rules without storing plaintext passwords.	Must	Test
FR-AUTH-003	The system shall require TOTP MFA before issuing a human portal session and shall limit challenge lifetime and failed attempts.	Must	Test
FR-AUTH-004	The system shall encrypt MFA secrets at rest and shall fail production startup when a development encryption key is configured.	Must	Inspection and Test
FR-AUTH-005	The system shall enforce role, plane, VPN entitlement and user IP or CIDR policy before granting privileged access.	Must	Test
FR-AUTH-006	The system shall issue HttpOnly session cookies with configured Secure and SameSite controls and shall revoke sessions after logout, password recovery or MFA reset.	Must	Test
FR-AUTH-007	The system shall protect state-changing browser requests with CSRF tokens and shall provide a CSRF-token endpoint for authenticated workflows.	Must	Test
FR-AUTH-008	The system shall record successful and failed authentication, password, MFA and recovery events in the audit trail.	Must	Inspection and Test

5.2 Data Request Governance

Requirement ID	Requirement	Priority	Verification
FR-REQ-001	External users shall be able to submit a data request containing purpose, required fields, time range, frequency and supporting justification.	Must	Demonstration
FR-REQ-002	The system shall assign a unique request identifier and status to every submission.	Must	Test
FR-REQ-003	Data Managers shall be able to review, comment on, approve or reject submitted requests.	Must	Demonstration
FR-REQ-004	The system shall preserve proposer, reviewer, timestamps, decisions and remarks as transaction history.	Must	Inspection
FR-REQ-005	A requester shall only view their own submissions unless a role grants wider	Must	Test

Requirement ID	Requirement	Priority	Verification
	governance access.		

5.3 Read Query Governance and Execution

Requirement ID	Requirement	Priority	Verification
FR-READ-001	The system shall accept only parameterised SELECT templates for governed read services.	Must	Test
FR-READ-002	Only parameter names beginning with :p_ shall be treated as input parameters; PostgreSQL format tokens and other colon text shall not be interpreted as parameters.	Must	Test
FR-READ-003	The SQL security filter shall reject DDL, DML, stacked statements, unsafe constructs and access outside the approved read policy.	Must	Test
FR-READ-004	Developers shall supply representative parameter values and perform a bounded dry run against an authorised sandbox Agent.	Must	Demonstration
FR-READ-005	A reviewer shall be able to edit SQL during review, rerun the dry run and submit changed SQL for approval with a recorded audit trail.	Must	Test
FR-READ-006	The system shall prohibit a proposer from approving their own query or material query revision.	Must	Test
FR-READ-007	Production execution shall use only approved query versions and the intersection of query-authorised, client-authorised and user-authorised Agents.	Must	Test
FR-READ-008	The system shall validate parameter types and bounds before dispatching a query to an Agent.	Must	Test
FR-READ-009	The Agent shall execute read queries with a read-only database identity and shall return bounded structured results and execution metadata.	Must	Test
FR-READ-010	Changes to approved SQL shall create a governed revision and shall not silently replace the approved definition.	Must	Inspection and Test

5.4 Reports and Exports

Requirement ID	Requirement	Priority	Verification
FR-RPT-001	Authorised users shall be able to discover and execute approved reports within their role, query and Agent scope.	Must	Demonstration
FR-RPT-002	The report viewer shall render returned columns and data without exposing SQL to unauthorised users.	Must	Test
FR-RPT-003	Permitted users shall be able to export report results to approved formats including CSV, Excel and PDF.	Should	Demonstration
FR-RPT-004	Export retrieval shall enforce ownership and authorisation checks and shall not permit cross-owner access.	Must	Test

Requirement ID	Requirement	Priority	Verification
FR-RPT-005	Report execution and export events shall be traceable through execution and audit records.	Must	Inspection

5.5 Agent Management

Requirement ID	Requirement	Priority	Verification
FR-AGT-001	Data Managers shall be able to register, activate, deactivate and inspect Agents without exposing database passwords in the UI.	Must	Test
FR-AGT-002	Each Agent shall represent one governed GePNIC database destination for mapping purposes.	Must	Inspection
FR-AGT-003	Read and write services may use the same Agent but shall retain independent permissions and verification states.	Must	Test
FR-AGT-004	Master-to-Agent governance and write calls shall use HTTPS and a dedicated inbound transit secret.	Must	Test
FR-AGT-005	The system shall expose Agent health and diagnostic status sufficient to distinguish routing, authentication, configuration and database failures.	Must	Demonstration
FR-AGT-006	A failure for one Agent operation shall identify the affected Agent and shall not display unrelated Agent rows as busy.	Must	Test

5.6 API Client Registry and Admission

Requirement ID	Requirement	Priority	Verification
FR-CLI-001	The system shall register API clients with a unique identifier, name, environment, API key, public key, IP allowlist and activation state.	Must	Test
FR-CLI-002	API keys shall identify clients but shall not by themselves grant query, write-service, Agent or data access.	Must	Test
FR-CLI-003	The registry shall analyse a submitted public key or certificate and display configuration and fingerprint status without receiving the private key.	Must	Demonstration
FR-CLI-004	The system shall enforce source IP or CIDR allowlists independently from cryptographic authentication.	Must	Test
FR-CLI-005	Read-query authorisation, read-Agent mapping, write-service mapping and write-service permission shall remain independently governed.	Must	Test
FR-CLI-006	The system shall allow controlled API-key rotation and shall audit the action without storing the new raw secret in logs.	Must	Test
FR-CLI-007	Versions and SDKs shall show only the integrations assigned to the selected client and shall fail closed for disabled or retired policies.	Must	Test
FR-CLI-008	UAT SDKs shall include the designated test identity and wallet settings; Production SDKs shall require the client-owned identity and public-key	Must	Inspection and Test

Requirement ID	Requirement	Priority	Verification
	registration.		
FR-CLI-009	SDK downloads shall include the contract-specific SDK, SHA-256 checksum, onboarding documentation and distribution email text with correct filenames.	Should	Inspection

5.7 Machine Request Security

Requirement ID	Requirement	Priority	Verification
FR-CRYPTO-001	Machine payloads shall be protected in transit using TLS and the approved cryptographic request envelope.	Must	Test
FR-CRYPTO-002	The request envelope shall use a fresh AES-256-GCM content key and IV, wrap the content key with the DOORS RSA public key and sign the exact plaintext bytes with the client private key.	Must	Test
FR-CRYPTO-003	The Master API shall authenticate the API key, source IP, registered public key and digital signature before executing the business operation.	Must	Test
FR-CRYPTO-004	Signed metadata shall bind the request identifier, issued time, client, operation, target Agents and parameters or payload.	Must	Test
FR-CRYPTO-005	The system shall reject stale, future-dated, malformed, downgraded or replayed requests with stable machine-readable error codes.	Must	Test
FR-CRYPTO-006	Where response encryption is required, DOORS shall encrypt the response for the registered client key and sign the response with the active DOORS signing key.	Must	Test
FR-CRYPTO-007	Swagger test flows shall require explicit identity selection, key password entry, payload entry or upload, encryption and execution before displaying the response.	Must	Demonstration

5.8 Write Service Governance

Requirement ID	Requirement	Priority	Verification
FR-WGOV-001	A proposer shall provide a service title, validated lowercase service code, purpose and representative JSON document.	Must	Test
FR-WGOV-002	The system shall analyse the JSON structure and propose readable database column names and PostgreSQL data types for selectable scalar fields.	Must	Demonstration
FR-WGOV-003	Column names, data types, required status and selected record-key fields shall remain editable or removable until approval.	Must	Test
FR-WGOV-004	The system shall preserve the complete received JSON as one payload value while treating extracted scalar columns as optional governed projections.	Must	Inspection
FR-WGOV-005	The proposer shall preview generated tables, views, functions and rules before generating the proposal.	Must	Demonstration

Requirement ID	Requirement	Priority	Verification
FR-WGOV-006	The system shall perform a sandbox write and read-back without leaving persistent sandbox objects or test records.	Must	Test
FR-WGOV-007	Write proposals and material revisions shall require independent approval and shall preserve proposer and reviewer transaction history.	Must	Test
FR-WGOV-008	Create New Revision shall copy the existing approved fields, types, required flags, record keys and service context for controlled editing.	Must	Demonstration
FR-WGOV-009	The approved definition shall have a stable hash that can be verified before applying a migration.	Must	Test
FR-WGOV-010	The write contract shall distinguish JSON contract version, transport protocol policy and stored record version.	Must	Inspection

5.9 Write Database Provisioning and Verification

Requirement ID	Requirement	Priority	Verification
FR-WDB-001	The system shall generate DBA SQL that creates or migrates the approved objects under the common doors_write schema.	Must	Inspection and Test
FR-WDB-002	The live API shall never create, alter or drop destination tables or schemas.	Must	Test
FR-WDB-003	Provisioning SQL shall be executed by the destination DBA and shall preserve existing data during approved migrations.	Must	Test
FR-WDB-004	The DBA shall own the schema, tables, views and SECURITY DEFINER write functions.	Must	Inspection
FR-WDB-005	The doors runtime role shall receive schema usage and approved function execution without table-creation, table-alteration or direct table-write privilege.	Must	Test
FR-WDB-006	The doors runtime role shall receive approved read access to the existing public schema, and the configured GePNIC application role shall receive read access to approved doors_write views.	Must	Test
FR-WDB-007	The downloadable SQL shall include explanatory DBA notes, prerequisite role handling, object creation, grants and verification queries.	Should	Inspection
FR-WDB-008	An approved write service shall allow multiple Agent mappings, with schema and permission verification performed independently for each Agent.	Must	Test
FR-WDB-009	Verification shall check schema objects, expected columns, primary keys, views, function signature, function owner, SECURITY DEFINER settings and runtime privileges.	Must	Test
FR-WDB-010	Verification failure shall return and log the exact failing check without disclosing credentials or secrets.	Must	Test

5.10 Write Request Processing and Versioning

Requirement ID	Requirement	Priority	Verification
FR-WRITE-001	An authorised client shall submit an approved JSON contract to its mapped and verified Agent destination.	Must	Test
FR-WRITE-002	The system shall derive the business record identity from the approved record-key fields and shall reject requests missing or invalidating those keys.	Must	Test
FR-WRITE-003	DOORS shall assign the next record version automatically; callers shall not be required to supply an expected record version.	Must	Test
FR-WRITE-004	The client shall supply or the SDK shall generate a UUID request identifier and shall reuse it only when retrying the exact same logical request.	Must	Test
FR-WRITE-005	A repeated request identifier with changed content shall return an idempotency conflict and shall not store data.	Must	Test
FR-WRITE-006	If the canonical received JSON is identical to the latest stored JSON for the business key, the system shall return UNCHANGED and shall not create a new record version.	Must	Test
FR-WRITE-007	If the canonical JSON differs and all validations pass, the system shall append one immutable snapshot and update the latest-record head atomically.	Must	Test
FR-WRITE-008	A client-wise configured version limit shall cap changed snapshots per business key; identical requests shall not consume the limit.	Must	Test
FR-WRITE-009	When a version limit is reached, the API shall return a clear conflict response instructing the caller to contact the DOORS Administrator.	Must	Test
FR-WRITE-010	The latest-record view shall expose the business key, current record version, relevant metadata and latest JSON payload without requiring consumers to scan history.	Must	Inspection and Test
FR-WRITE-011	For a single-instance Agent database, GePNIC consumers shall be able to locate the latest record by approved business keys; instance_id shall remain available for consolidated or multi-instance use.	Must	Test
FR-WRITE-012	Requests with missing required fields or incompatible approved types shall be rejected before the database write function is invoked.	Must	Test
FR-WRITE-013	Unknown JSON fields shall follow the configured contract-evolution policy and shall never silently become extracted database columns without approval.	Must	Test
FR-WRITE-014	Write responses shall identify outcome, replay status, request ID, destination instance and current record version.	Must	Test

5.11 Document Services

Requirement ID	Requirement	Priority	Verification
FR-DOC-001	The system shall discover eligible documents using an approved manifest query and configured document-service policy.	Must	Test
FR-DOC-002	Before streaming bytes, the Agent shall evaluate download eligibility using the	Must	Test

Requirement ID	Requirement	Priority	Verification
	approved query or database function.		
FR-DOC-003	The system shall stream approved files without loading the complete file into Master API memory.	Must	Analysis and Test
FR-DOC-004	The service shall prevent path traversal and access outside configured document roots.	Must	Test
FR-DOC-005	The Master API shall compute and return the source SHA-256 checksum while streaming.	Must	Test
FR-DOC-006	The SDK shall write to a temporary part file, verify the downloaded checksum and rename only after integrity succeeds.	Must	Test
FR-DOC-007	The client shall submit a download receipt containing source and destination checksums and correlation information.	Must	Test
FR-DOC-008	Document queries, denials, downloads, failures and receipts shall be audited without storing document content in logs.	Must	Inspection

5.12 Subscriptions, Usage and Access Scheduling

Requirement ID	Requirement	Priority	Verification
FR-SUB-001	Data Managers shall be able to configure API-user subscription periods, product/service status, Agent association, accounting metadata and notices.	Must	Demonstration
FR-SUB-002	Manual service suspension shall deny applicable machine requests independently of advisory subscription dates.	Must	Test
FR-SUB-003	Weekly schedules and date-specific access periods shall restrict applicable machine requests using inclusive starts and exclusive ends.	Must	Test
FR-SUB-004	Where multiple governing schedules apply, the request shall be permitted only when every applicable schedule permits it.	Must	Test
FR-SUB-005	The system shall meter completed application response-body bytes, status, endpoint, elapsed time and record counts without buffering streamed documents.	Must	Test
FR-SUB-006	API users shall see only their assigned clients, subscription information, usage and notices.	Must	Test
FR-SUB-007	The system shall preserve subscription and accounting history and shall not expose destructive deletion of accounting periods.	Must	Inspection
FR-SUB-008	Interactive report permissions shall remain independent from machine-service subscription gates.	Must	Test

5.13 Audit, Execution Logs and Diagnostics

Requirement ID	Requirement	Priority	Verification

Requirement ID	Requirement	Priority	Verification
FR-AUD-001	Every governed state change and security-relevant event shall create an immutable audit record containing actor, action, target, timestamp, result and trace identifier.	Must	Inspection and Test
FR-AUD-002	Execution logs shall correlate client, API key used, source IP, endpoint, Agent, parameters or decrypted business request, result, duration and trace ID as permitted by log-access policy.	Must	Inspection
FR-AUD-003	Private keys, passwords, wallet secrets, transient AES keys and raw authentication credentials shall never be written to logs.	Must	Test
FR-AUD-004	Authorised Data Managers shall be able to search logs by trace ID, client, API key, endpoint, Agent, status and time range.	Must	Demonstration
FR-AUD-005	Write-result-unknown conditions shall log the downstream failure stage and cause while returning a safe retry instruction to the caller.	Must	Test
FR-AUD-006	Operational error responses shall use stable codes, safe messages and trace IDs without exposing stack traces or database credentials.	Must	Test

5.14 AIra Assistance

Requirement ID	Requirement	Priority	Verification
FR-AIRA-001	AIra shall answer DOORS feature and operational questions using content retrieved from the governed vector knowledge index.	Must	Test
FR-AIRA-002	AIra shall rank and return only relevant knowledge topics and shall not append unrelated retrieved sections to an answer.	Must	Test
FR-AIRA-003	Operational questions requiring current counts or activity shall use authorised live data tools rather than static documentation.	Must	Test
FR-AIRA-004	AIra answers shall identify the supporting source title or operational data source.	Should	Inspection
FR-AIRA-005	Administrators shall be able to reindex approved knowledge after a documented knowledge update.	Must	Demonstration
FR-AIRA-006	Knowledge source versions and checksums shall be controlled so that the active vector index can be reconciled with approved source documents.	Should	Inspection
FR-AIRA-007	AIra shall enforce the caller's role and data scope for every live operational tool call.	Must	Test
FR-AIRA-008	AIra shall decline unsupported or unauthorised questions rather than fabricate platform state.	Must	Test

6 External Interface Requirements

6.1 User Interface

- The UI shall be responsive, role-aware and usable on supported desktop browsers, with readable text contrast and clear disabled-state guidance.
- Multi-stage governance forms shall show a consistent roadmap, current stage, mandatory-field markers, next action and contextual help.
- Actions shall use distinct, descriptive buttons with icons where they improve recognition; one form shall not expose redundant buttons that open the same operation.
- Opening a proposal, approval, mapping or client workflow shall use a clear modal, drawer or dedicated view rather than inserting an unnoticed form below a grid.
- The footer shall display the deployed product version and make release notes available.

6.2 API Interface

Interface	Requirement
Interactive APIs	Cookie-authenticated JSON over HTTPS with CSRF protection for state-changing calls.
Machine APIs	API-key and asymmetric-key authenticated HTTPS requests with approved encrypted payload envelope.
OpenAPI	Contract-specific schemas shall include approved field types, required fields, constraints and examples.
SDK	Generated Java client shall implement the approved signing, encryption, truststore, retry and response-processing rules.
Agent	Master-to-Agent endpoints shall authenticate the Master transit secret, validate target profiles and return bounded structured diagnostics.
Errors	JSON problem details or stable service response containing HTTP status, code, safe message and trace ID.

6.3 Database Interface

Agents shall use named, environment-specific JDBC target profiles. Read operations shall use a read-only role. Write operations shall invoke only the approved SECURITY DEFINER function through the doors runtime role. Database passwords, where required, shall be supplied through protected runtime configuration and not committed to source control.

6.4 Network and Proxy Interface

All non-local communication shall use HTTPS. Reverse proxies shall preserve the required route prefix, client address evidence and approved headers; restrict methods and source networks; and expose only routes assigned to the selected plane. Health and governance paths shall be configured explicitly rather than inferred from read-query routes.

7 Data Requirements

7.1 Core Data Domains

Domain	Representative records	Retention characteristic
Identity	Users, roles, MFA state, VPN/IP entitlements and sessions	Security-policy controlled
Governance	Requests, query revisions, write contracts, approvals, mappings and policies	Historical and auditable
API clients	Client identifiers, hashed or protected keys, public keys, IP ranges and activation	Retain change history
Operations	Execution logs, errors, traces, metrics and Agent health	Operational retention policy
Documents	Manifest metadata, download audits and receipts	No document body in Master database
Write records	Immutable JSON snapshots, extracted fields, head state, receipts and permissions	Business-policy controlled
AIra	Approved knowledge metadata, chunks, embeddings and retrieval evidence	Versioned and reindexable

7.2 Data Integrity

- Primary and foreign keys shall protect governance relationships and prevent orphan mappings.
- Write snapshots shall be immutable; head updates and snapshot insertion shall be atomic.
- Request receipts shall enforce idempotency at the database boundary.
- Approval status transitions shall be validated server-side and shall not rely on UI state.
- Dates and instants shall use explicit timezone rules; operational schedules shall use Asia/Kolkata unless a contract states otherwise.
- JSON comparison for unchanged detection shall use a canonical semantic representation rather than textual whitespace or key order.

7.3 Data Retention and Disposal

Retention periods for audit, execution, subscription, document and write-history records shall be approved by the programme owner and security authority. Disposal shall be authorised, logged and consistent with legal-hold requirements. The system shall not expose bulk deletion of governance or accounting history through routine UI operations.

8 Non Functional Requirements

8.1 Security

Requirement ID	Requirement	Priority	Verification
NFR-SEC-001	The system shall apply deny-by-default authorisation at plane, role, client, service, Agent and data-scope boundaries.	Must	Security Test
NFR-SEC-002	TLS shall protect all remote browser, machine and Master-to-Agent traffic using	Must	Inspection and

Requirement ID	Requirement	Priority	Verification
	approved certificates and protocols.		Test
NFR-SEC-003	Passwords, MFA secrets, API keys and private cryptographic material shall be stored or handled according to their sensitivity and shall never be exposed in routine UI responses.	Must	Security Test
NFR-SEC-004	Input validation shall protect SQL, JSON, filenames, paths, headers and rendered text against injection and traversal attacks.	Must	Security Test
NFR-SEC-005	Rate limiting shall protect interactive and machine endpoints and shall return HTTP 429 with safe retry guidance when exceeded.	Must	Load and Security Test
NFR-SEC-006	Security headers, content security policy and secure cookie settings shall be configured for each web plane.	Must	Security Test
NFR-SEC-007	Cryptographic algorithms and key sizes shall be centrally configurable and replaceable through a governed migration.	Should	Inspection
NFR-SEC-008	Production configurations shall reject known development secrets, trust-all TLS modes and insecure fallback credentials.	Must	Test

8.2 Performance and Capacity

Requirement ID	Requirement	Priority	Verification
NFR-PERF-001	Interactive operations shall provide visible progress and complete within the approved service-level target under expected load.	Must	Performance Test
NFR-PERF-002	Read queries shall use server-side timeouts, bounded result sizes and configurable row limits.	Must	Test
NFR-PERF-003	Document streaming shall use bounded memory independent of document size.	Must	Performance Test
NFR-PERF-004	The system shall support horizontal growth of stateless web and Master API instances where shared persistence and idempotency controls are configured.	Should	Analysis
NFR-PERF-005	AIra retrieval shall bound the number and size of knowledge chunks supplied for answer generation.	Must	Performance Test

8.3 Availability and Reliability

Requirement ID	Requirement	Priority	Verification
NFR-REL-001	Failure of one Agent shall not corrupt governance state or prevent use of unrelated healthy Agents.	Must	Fault Test
NFR-REL-002	Write timeouts with uncertain downstream results shall be safe to retry using the same request identifier and exact payload.	Must	Fault Test
NFR-REL-003	Database state transitions requiring consistency shall execute transactionally and roll back on failure.	Must	Test

Requirement ID	Requirement	Priority	Verification
NFR-REL-004	Services shall expose health information suitable for monitoring without exposing secrets.	Must	Demonstration
NFR-REL-005	Backup and recovery procedures shall cover the Master database, configuration, keys, approved knowledge sources and destination write schema.	Must	Recovery Test

8.4 Auditability and Observability

Requirement ID	Requirement	Priority	Verification
NFR-OBS-001	Every request crossing a service boundary shall carry or receive a trace identifier that is preserved in logs and safe error responses.	Must	Test
NFR-OBS-002	Logs shall distinguish validation, policy, proxy, Agent, database, cryptographic and timeout failures.	Must	Inspection
NFR-OBS-003	Operational metrics shall support service health, error rate, latency, egress and Agent availability monitoring.	Should	Demonstration
NFR-OBS-004	System clocks shall be synchronised and audit timestamps shall be unambiguous.	Must	Inspection

8.5 Usability and Accessibility

Requirement ID	Requirement	Priority	Verification
NFR-UX-001	Normal body text, labels, validation messages and table actions shall remain readable at supported browser zoom levels.	Must	Visual Test
NFR-UX-002	Mandatory inputs shall be marked consistently and validation shall identify the actual invalid field and correction.	Must	Visual Test
NFR-UX-003	Keyboard navigation, focus order, labels and colour contrast shall meet the approved accessibility baseline.	Should	Accessibility Test
NFR-UX-004	Operational workflows shall provide step-specific guidance without unrelated explanatory content.	Must	Usability Test
NFR-UX-005	Downloadable guides and PDFs shall preserve selectable and copyable commands where technically possible.	Should	Inspection

8.6 Maintainability and Portability

Requirement ID	Requirement	Priority	Verification
NFR-MNT-001	UI, Master API and Agent API shall be independently buildable and deployable with versioned artefacts and SHA-256 checksums.	Must	Build Test

Requirement ID	Requirement	Priority	Verification
NFR-MNT-002	Environment-specific URLs, target profiles, secrets and truststores shall be externalised from application binaries.	Must	Inspection
NFR-MNT-003	Database schema changes shall use controlled, repeatable migrations that preserve previously applied migration checksums.	Must	Migration Test
NFR-MNT-004	Generated SDKs and OpenAPI documents shall be derived from the approved service contract to prevent drift.	Must	Test
NFR-MNT-005	Automated tests shall cover policy, cryptography, governance transitions, SQL filtering, write versioning and regression of unaffected read services.	Must	Inspection

9 Business Rules and Governance

Rule ID	Business rule
BR-001	A proposer shall not approve their own query, write contract or material revision.
BR-002	API-key possession alone shall never grant business-service access.
BR-003	Read and write permissions are independent even when the same API client and Agent are used.
BR-004	A live write call invokes an existing approved database function and never creates schema objects.
BR-005	Each Agent maps to one governed GePNIC database; multiple Agents may be mapped to the same approved service.
BR-006	An identical latest JSON payload does not create a new stored record version.
BR-007	A request ID may be replayed only with the exact same logical request content.
BR-008	New extracted columns or changed types require a governed contract revision and corresponding DBA migration.
BR-009	Unknown JSON fields follow the active contract-evolution policy but never become database projections automatically.
BR-010	The vector database is the runtime retrieval source for Alra; approved source documents remain the authoritative knowledge baseline.
BR-011	Private keys remain under the control of the calling application owner.
BR-012	All privilege, approval and policy changes are auditable and attributable to an authenticated actor.

10 Deployment and Operational Requirements

10.1 Deployment Units

- DOORS UI production distribution, deployed under the correct environment-specific base path.
- Master API executable JAR with external application configuration, truststores, identities and database connectivity.

- Agent API executable JAR with retained environment-specific target, sandbox, transit-secret and database configuration.
- Reverse-proxy configuration for Admin, External, API and Agent routes.
- Approved database migrations and service-specific DBA provisioning or migration SQL.
- SHA-256 manifests for deployable artefacts and an approved deployment record.

10.2 Deployment Sequence

1. Back up current binaries, UI distribution, configuration and databases according to the rollback plan.
2. Validate and apply approved Master database migrations before starting a build that depends on them.
3. Deploy the Master API and preserve external secrets and environment-specific configuration.
4. Deploy the Agent API when Agent-side routes, validation, verification or write behaviour changed.
5. Deploy the UI built with the target environment base paths and invalidate stale browser/proxy caches.
6. Restart or reload services through the approved service manager and verify health and proxy routing.
7. Reindex AIra only when approved knowledge sources changed, then run grounded-answer smoke tests.
8. Execute read, write, document, audit and unaffected-service smoke tests and record the results.

10.3 Configuration Management

Secrets shall be supplied by protected environment variables, secret stores or permission-restricted configuration. Deployments shall not overwrite staging or production configuration with local development files. UI builds shall be produced with the intended environment profile; incorrect base paths shall be detected by a deployment smoke test of the CSRF endpoint.

11 Acceptance and Traceability

11.1 Acceptance Gates

Gate	Minimum evidence
Build	UI type check and production build; Master and Agent unit/integration tests; reproducible packages and checksums.
Security	Authentication, RBAC, plane routing, CSRF, input validation, cryptography, IP policy, replay and negative tests.
Governance	Independent approval, revision history, audit records and invalid-transition tests.
Read services	Parameter parsing, SQL security, dry run, mapped execution, result and unaffected-read regression.
Write services	Contract analysis, sandbox, approval, DBA SQL, Agent verification, STORED, UNCHANGED, replay and limit tests.
Documents	Eligibility allow/deny, streaming, path security, checksum and receipt tests.
Operations	Proxy routes, health, logging, trace correlation, backup and rollback rehearsal.
AIra	Relevant feature answers, workflow answers, live activity questions, source attribution and unauthorised-scope denial.

11.2 Requirement Traceability

Every test case shall reference one or more requirement IDs from this SRS. Every Must requirement shall have at least one positive acceptance test and, where a security or validation boundary exists, at least one negative test. Defects and change requests shall reference the affected requirement, implemented version and verification evidence.

SRS group	Primary verification suite
FR-AUTH and NFR-SEC	Authentication, role matrix, plane routing, CSRF and security test suites
FR-READ and FR-RPT	SQL filter, parameter parser, query lifecycle, Agent and report tests
FR-WGOV, FR-WDB and FR-WRITE	Write governance, SQL generation, Agent verification, function and SDK tests
FR-DOC	Document policy, traversal, streaming, checksum and receipt tests
FR-SUB	Subscription, schedule, notice and egress accounting tests
FR-AUD and NFR-OBS	Audit persistence, redaction, search and trace-correlation tests
FR-AIRA	Retrieval relevance, workflow, live activity, source and authorisation tests

Appendix A Status Models

Domain	Representative statuses	Transition constraint
Data request	Pending, Approved, Rejected	Decision requires authorised reviewer and recorded remarks.
Query	Draft or Pending, Tested, Approved, Rejected, Disabled	Approval requires successful dry run; material SQL change returns to review.
Write proposal	Draft, Sandbox Tested, Pending Approval, Approved, Rejected, Disabled	DBA installation and Agent verification occur after approval.
Agent mapping	Verification Pending, Verified, Failed	Each mapped Agent is verified independently.
API client	Inactive, Active	Activation does not replace query/service/Agent permissions.
Write result	Stored, Unchanged, Replayed, Rejected, Result Unknown	Outcome controls versioning and retry behaviour.

Appendix B Error and Response Behaviour

HTTP status	Representative meaning	Client behaviour
200	Successful read or unchanged/replayed write	Use returned outcome and record version.
201	New write version stored	Record request ID and returned version.

HTTP status	Representative meaning	Client behaviour
400 or 422	Malformed request, invalid contract, type or parameter	Correct the request; do not retry unchanged.
401	Authentication missing or expired	Authenticate again or correct machine credentials.
403	Role, IP, signature, service, Agent or business-scope denial	Use code and trace ID; request authorised correction.
404	Unknown or unavailable governed resource or incorrect proxy route	Check route, deployment and registry state.
409	Workflow conflict, replay conflict, idempotency conflict or version limit	Follow the specific code; never change payload while reusing request ID.
421	Unknown or incorrect request plane	Use the assigned hostname and route.
429	Rate limit exceeded	Wait for the allowed window and use backoff.
502 or 503	Agent or dependency unavailable	Use trace ID and retry only when marked retryable.
504	Outcome uncertain after timeout	Retry the exact request with the same request ID.

Appendix C Glossary

Term	Meaning
Agent	DOORS service deployed near a GePNIC node that executes governed database and document operations.
AIra	DOORS assistant that combines retrieval-grounded platform knowledge with authorised live operational tools.
Business key or record key	Approved JSON fields that uniquely identify the versioned business record.
Contract version	Version of an approved JSON structure, required fields, types, record keys and projections.
DOORS	Data Orchestration and On Demand Reporting System.
GePNIC	Government e Procurement System of National Informatics Centre.
Idempotency	Property ensuring an exact retry does not create an unintended duplicate business effect.
Instance ID	Destination context retained in write records for multi-instance or consolidated use.
Plane	Isolated Admin, External or API request surface with a dedicated hostname and route policy.
Record version	Automatically assigned immutable snapshot number for one business key.
SDK	Generated software development kit implementing the approved API contract and security protocol.
SoD	Separation of duties between proposing, approving, administering and consuming.
Vector index	Searchable embedding index used to retrieve relevant approved knowledge for AIra.